Comment on the collision-resistance proof of the TaiChi IFS mode

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Abstract

The TaiChi v2.0 collision claim is $\lambda_{\text{coll}} = \min\{d/2,\, c\}$, i.e. $T_{\text{coll}} \approx \min\{2^{d/2}, 2^c\}$, stated for general parameters. We show this is not correct. There is a structural inverse-rate collision route, built on the cumulative-XOR relation in the capacity, that costs about $2^{3c/4}$ queries; it is below $\min\{2^{d/2}, 2^c\}$ whenever $d > 3c/2$. We describe the route, report that we implemented and verified it against the submitted permutation P_{1920}, and locate the exact step in the completeness argument that misses it. The three submitted instances have $c = d$, which is below the $d > 3c/2$ threshold, so their $2^{d/2}$ collision level is unaffected and the parameters are not broken; the error is in the general claim and proof. We also note that second-preimage resistance is asserted but not proven; we have neither a proof of the 2^c level nor an attack below it, so its status is open.

Disclaimer. The generic collision attack described in this note was produced by the author. It was subsequently verified with the assistance of AI, and all attack code (the reimplementation of P_{1920} and its inverse, the attack at configurable c, the scaling measurement, and the independent verifier) was written by AI. This note was itself drafted in its entirety by AI.

1 Summary

TaiChi [1] is a hash family submitted to the NGCC program, built from an interleaved two-permutation sponge (IFS) mode over a 1920-bit permutation P_{1920}, with fixed-output instances TaiChi-512, TaiChi-768 and TaiChi-1024. Let the IFS state be (S, L, R) with $|S| = r$ and $|L| = |R| = c$, and let $P_1 = P_{1920}(\cdot, 1)$, $P_2 = P_{1920}(\cdot, 2)$. The claimed collision level (Table 2.1) is $\min\{2^{d/2}, 2^c\}$. The true collision resistance of the mode is at most

$$\min\{2^{d/2},\, 2^{3c/4}\},$$

so the claim overstates it for $d > 3c/2$. The $2^{3c/4}$ route is a real attack, not a bound artifact; we ran it against P_{1920}. For the submitted instances $d = c < 3c/2$, the $2^{d/2}$ digest birthday is the binding bound, so the stated $2^{c/2}$ collision level holds and no submitted instance is broken. The issue is that the collision theorem is stated and proved for general (r, c, d), and the general statement is wrong.

2 The mode and the claim

The step function is

$$(A, B) = P_1\big((S \oplus M) \,\|\, (L \oplus \text{enc}(\mathsf{D}))\big), \quad (S', C) = P_2\big(A \,\|\, (R \oplus B)\big), \quad (S, L, R) \leftarrow (S',\, C,\, B \oplus C),$$

with $\text{enc}(\tau_{\text{Ab}}) = 0^{c-2}\|01$, $\text{enc}(\tau_{\text{Fin}}) = 0^{c-2}\|10$, IV $= (0, 0, 0)$, 10^*1 padding, and digest the first d bits of the final rate word S (a squeeze is used only when $d > r$). The collision bound

(Theorem 1) is

$$\mathrm{Adv}^{\mathrm{coll}} \le O\Big(\tfrac{q_{\mathrm{eval}}^2}{2^d}\Big) + O\Big(\tfrac{q_{\mathrm{eval}} q_P}{2^c}\Big) + O\Big(\tfrac{\sigma^2}{2^{2c}}\Big) + O\Big(\tfrac{\sigma^2 + \sigma q_P}{2^n}\Big) + O\Big(\tfrac{q_s^2}{2^{r+2c}}\Big),$$

and the specification reads it as $T_{\mathrm{coll}} \approx \min\{2^{d/2}, 2^c\}$: the $2^{d/2}$ digest birthday and a "2^c capacity cap" from the $q_{\mathrm{eval}} q_P / 2^c$ term.

3 A collision route at $2^{3c/4}$

Process a two-block message (m_0, m_1) from the all-zero IV. The first (absorbing) block gives

$$P_1(m_0 \,\|\, \tau_{\mathrm{Ab}}) = (a, x), \qquad P_2(a \,\|\, x) = (s, y),$$

since $R_0 = 0$, so the state after block one is $(S_1, L_1, R_1) = (s,\, y,\, x \oplus y)$. The second register has accumulated $R_1 = x \oplus y$. Fix an arbitrary rate value $A^\star$. For q values z compute the inverse query $P_1^{-1}(A^\star \,\|\, z) = (u, \ell)$. Whenever a first-block root has $y = \ell \oplus \tau_{\mathrm{Fin}}$, set $m_1 = s \oplus u$; then the final P_1 input is $(s \oplus m_1) \,\|\, (y \oplus \tau_{\mathrm{Fin}}) = u \,\|\, \ell$, with output $A^\star \,\|\, z$, and the final P_2 input is

$$A^\star \,\|\, (R_1 \oplus z) = A^\star \,\|\, (x \oplus y \oplus z) = A^\star \,\|\, D, \qquad D = x \oplus y \oplus z.$$

Two such continuations with equal D feed the same final P_2 call, so they produce the same final rate word S (and the same L), hence the same digest. With q forward roots and q inverse queries the join on the c-bit value y yields about $N \approx q^2/2^{c+2}$ continuations (the padding condition costs a constant factor); a birthday on the c-bit D among them needs $N \approx 2^{c/2}$, so

$$q \approx 2^{3c/4}.$$

We implemented P_{1920} and its inverse (reproducing the four reference TaiChi-512 digests bit for bit), ran the attack on reduced-capacity variants with the real permutation, and measured the first-collision query count. The exponent is 0.7475 ± 0.0015 over $c \in \{8, 16, 24\}$ for P_{1920} and 0.7499 ± 0.0011 over $c \in \{12, \ldots, 28\}$ for an unrelated permutation; every collision is checked by re-hashing both messages through the reference construction. As a larger check, a full collision at $c = 40$ (target scale 2^{30}) was produced against the real P_{1920} and verified through the reference construction; it took about fifteen minutes on a laptop with eight cores and 8 GB of RAM.

4 Where the proof misses it

The completeness lemma (Lemma 5) states that under $\neg(\mathsf{Bad_{fw}} \vee \mathsf{Bad_{hp}} \vee \mathsf{Bad_{sq}})$ a collision implies ProjColl or PrimTargetColl. The route above is none of these cleanly.

- It is not a state merge. The two colliding messages share (S, L) but differ in R, indeed $R \oplus R' = z \oplus z'$. So $\mathsf{Bad_{hp}}$ (a $2c$-bit hidden-pair collision) does not occur; the route needs no state merge.
- It is not a fresh-output coincidence. The two final P_1 outputs share the r-bit rate part $A^\star$, but that alignment is produced by the adversary's own inverse queries, where the output $A^\star \,\|\, z$ is chosen. Nothing is freshly sampled onto an existing value, so $\mathsf{Bad_{fw}}$ (a full n-bit coincidence, charged 2^{-n}) does not fire. The coincidence that matters is the equality of two final P_2 inputs $A^\star \,\|\, D$, and with the rate pre-aligned to $A^\star$ this is a c-bit event in D, not an n-bit event.
- It is not the random digest event ProjColl (the full S is forced equal, not matched on a fresh d-bit projection), and it is not PrimTargetColl (there is no already-fixed target digest; both digests are free).

The "structured P_1-input replay" Remark considers forcing the same P_1 *input* $I_i^1 = I_j^1$, and correctly notes that two states then merge only when $R_i = R_j$. The route here forces the same P_1 *output rate* $A^\star$ through inversion, across distinct P_1 inputs, and it collides the output without a state merge. That case is not covered.

The only term large enough to absorb the route formally is $q_{\text{eval}} q_P / 2^c$, and reading it as a 2^c cap assumes a single fixed target ($q_{\text{eval}} = O(1)$), which is the preimage setting. For a collision the adversary forms about $2^{c/2}$ candidates, both counts scale with the query budget, and the term becomes a $T^2/2^c$ birthday that supports only a $2^{c/2}$ guarantee. The realizable attack sits at $2^{3c/4}$, between that guarantee and the claimed 2^c.

Relation to the San Mai attack. The route is an instance of the Universal San Mai Attack formalized by Li, Li, Hu and Liu [2]: a meet-in-the-middle that sandwiches inverse permutation queries between forward ones to exploit a linear capacity feedforward. In their classification of generalized (porifera) sponges, single-feedforward designs such as JH and FP fall to $b/3$ and the double sponge to $2c/3$ under this attack; the TaiChi IFS mode is a two-permutation feedforward sponge, and the $2^{3c/4}$ collision above is its instance. The same work isolates the construction that resists this family up to about 2^c, named VFB, which takes rate at least capacity, feeds forward (A, M) rather than the previous capacity, and uses a distinct finalization. A TaiChi proof that aims to be correct for general (d, c) has to account for this attack family.

5 When the claim fails, and why the submitted instances survive

The route collides the entire rate word S, so it is a digest collision whenever $d \le r$ (no squeeze). It beats the claim when $2^{3c/4} < \min\{2^{d/2}, 2^c\}$, i.e. when $d > 3c/2$ (and $d \le r$, which holds for $c < 768$). In that regime the specification claims $2^{d/2}$ or 2^c but the real cost is $2^{3c/4}$.

The three submitted instances take $c = d$, hence $d = c < 3c/2$. There the digest birthday $2^{d/2} = 2^{c/2}$ is cheaper than $2^{3c/4}$, so it is the binding bound and the claimed $2^{c/2}$ collision level is correct. No submitted instance is broken by this route. The defect is in the general theorem: it is stated and proved without assuming $c = d$, and it is false for $d > 3c/2$. A corrected claim is $T_{\text{coll}} \approx \min\{2^{d/2}, 2^{3c/4}\}$.

6 Second preimage

Section 2.2.7 states that second-preimage resistance is not proven in the current version and is deferred to future work, with only the generic 2^c upper bound asserted. We add nothing to it: we have neither a proof of the 2^c level nor an attack below it. We did look for an improvement and did not find one. The $2^{3c/4}$ collision idea does not transfer, because it relies on a birthday between two free messages, whereas a second preimage must hit the one fixed value of the target and so needs about 2^c candidates rather than $2^{c/2}$. A long-message splice does not help either: landing on an internal state of the target fixes both capacity registers, a $2c$-bit match, and the inverse-rate trick removes the r-bit rate but neither register, so the splice costs about $2^{2c}/L$ against a target of L blocks and bottoms out near 2^c. These are the absence of an attack, not a lower bound. The status of second-preimage resistance is therefore open: unproven, and, as far as we found, not broken.

7 Recommendation

1. Correct the collision claim to $\min\{2^{d/2}, 2^{3c/4}\}$, and revise the completeness argument to account for rate-aligned output collisions (two final P_2 inputs made equal through inverse

P_1 queries, colliding on the c-bit capacity without a state merge).

2. If any future instance uses $d > 3c/2$, or a longer digest or XOF output than the current fixed outputs, the $2^{3c/4}$ route becomes the binding collision bound and must be reflected in the parameters.
3. For a finalization that provably resists this attack family, consider the VFB construction of [2] (rate at least capacity, an (A, M) feedforward, and a distinct chop), which is proven indifferentiable up to about 2^c.

Reproducibility. A reference-matching implementation of P_{1920} (forward and inverse) and of the mode, the inverse-rate collision attack at configurable c, the scaling measurement, and an independent verifier are available on request. The implementation reproduces the reference TaiChi-512 digests, and every reported collision is verified by re-hashing both messages through the reference construction.

Scope. This is a mode-level comment in the independent ideal-permutation model used by the specification. It makes no claim about the concrete permutation P_{1920}, and it does not dispute the preimage bound.

References

[1] Zhongyi Zhang, Xiaobin Yu, Meicheng Liu, Guoyan Zhang, Shiyue Diao, and Peng Nie. *TaiChi*, version 2.0. Submission to the Next-generation Commercial Cryptographic Algorithms Program (NGCC), Institute of Commercial Cryptography Standards (ICCS), 30 June 2026.

[2] Hongli Li, Changlun Li, Honggang Hu, and Fengmei Liu. *Indifferentiability of the Sponge Hash Family: New Attacks and the Optimal Construction*. IACR Transactions on Symmetric Cryptology, 2026(1):31–75. DOI:10.46586/tosc.v2026.i1.31-75.